



Computer Network Headers

Quick Reference & Cram Sheet

ICMP • IPv4 • IPv6 • TCP • UDP • Ethernet • Token Ring • SSL/TLS

🔑 **Key:** Highlighted + Underlined = Critical field to memorize

► ICMP — Internet Control Message Protocol (Network Layer)

💡 ICMP is used for error reporting and diagnostics (ping, traceroute). It rides inside an IP packet. No ports — it's not for data transfer!

Field	Size (bits)	Purpose / What it does
<u>Type</u>	8	Identifies the ICMP message category (e.g., 0=Echo Reply, 8=Echo Request, 3=Dest Unreachable, 11=Time Exceeded)
<u>Code</u>	8	Sub-type within the Type field. Provides more detail (e.g., Type 3 Code 0 = Net Unreachable, Code 1 = Host Unreachable).
Checksum	16	Error-detection over the ICMP header + data. Receiver recalculates and discards if mismatch.
Rest of Header	32	Variable — depends on Type/Code. For Echo: Identifier (16 bits) + Sequence Number (16 bits).
Data	Variable	Optional payload (e.g., timestamp or padding in ping). Not present in all ICMP types.

Common Type values to memorize: 0=Echo Reply 8=Echo Request 3=Dest Unreachable 11=Time Exceeded (TTL=0 → traceroute!)

► IPv4 Header — Internet Protocol Version 4 (Network Layer)

💡 IPv4 header is 20 bytes minimum (up to 60 bytes with options). It provides addressing and fragmentation for packets across networks.

Field	Size (bits)	Purpose / What it does
<u>Version</u>	4	IP version. Always 0100 (= 4) for IPv4. Tells the receiver which protocol format to expect.
<u>IHL (Header Length)</u>	4	Internet Header Length — number of 32-bit words in the header. Minimum = 5 (→ 20 bytes). Needed because Options field is variable.
DSCP / ToS	8	Differentiated Services Code Point — QoS priority marking (e.g., VoIP gets higher priority than email).
<u>Total Length</u>	16	Total size of IP datagram in bytes (header + data). Max = 65,535 bytes.
<u>Identification</u>	16	Unique ID for a datagram. All fragments of the same original packet share this ID.
<u>Flags</u>	3	Bit 0: reserved. Bit 1: DF (Don't Fragment). Bit 2: MF (More Fragments — set on all but last fragment).
<u>Fragment Offset</u>	13	Byte offset of this fragment relative to original datagram (in units of 8 bytes). Used to reassemble fragments.
<u>TTL (Time to Live)</u>	8	Decrement by 1 at each router. Packet dropped when TTL=0 → prevents infinite loops. Traceroute exploits this!
<u>Protocol</u>	8	Identifies the transport layer protocol inside. TCP=6, UDP=17, ICMP=1, OSPF=89.
Header Checksum	16	Error check on the IP header only (not data). Recalculated at every router since TTL changes.
<u>Source IP Address</u>	32	Sender's IPv4 address (e.g., 192.168.1.10). 4 octets, dotted decimal notation.
<u>Destination IP Address</u>	32	Receiver's IPv4 address. Routers use this to forward the packet hop-by-hop.
Options	0–320	Optional fields for security , routing , timestamps . Present only if IHL > 5. Rarely used in practice.

► IPv6 Header — Internet Protocol Version 6 (Network Layer)

💡 IPv6 has a fixed 40-byte header (simpler than IPv4!). No fragmentation at routers, no checksum, no options — replaced by Extension Headers.

Field	Size (bits)	Purpose / What it does
<u>Version</u>	4	Always 0110 (= 6) for IPv6. First thing checked to distinguish IPv4 vs IPv6.
Traffic Class	8	Similar to IPv4 DSCP — QoS and priority marking for differentiated services.
<u>Flow Label</u>	20	Marks packets belonging to the same flow (e.g., a video stream). Routers can treat them consistently without re-reading headers.
<u>Payload Length</u>	16	Length of the data following the IPv6 header (including extension headers). Unlike IPv4, does NOT include the fixed header.
<u>Next Header</u>	8	Identifies what follows this header — either a transport protocol (TCP=6, UDP=17) or an extension header. This replaces IPv4 Protocol+Options.
<u>Hop Limit</u>	8	Same role as IPv4 TTL. Decrement by 1 per router; packet dropped at 0.
<u>Source Address</u>	128	Sender's IPv6 address. 128 bits = 16 bytes. Written in 8 groups of 4 hex digits (e.g., 2001:0db8::1).
<u>Destination Address</u>	128	Receiver's IPv6 address. Routers forward based on this.

IPv4 vs IPv6 at a glance: Header size 20B→40B | TTL→Hop Limit | No checksum in v6 | No fragmentation at routers | 32-bit addr→128-bit | Options→Extension Headers

► TCP Header — Transmission Control Protocol (Transport Layer)

💡 TCP is connection-oriented, reliable, ordered delivery. Uses 3-way handshake (SYN → SYN-ACK → ACK). Minimum header is 20 bytes.

Field	Size (bits)	Purpose / What it does
<u>Source Port</u>	16	Port number of the sender application (e.g., ephemeral port 49152–65535). Identifies the sending process.
<u>Destination Port</u>	16	Port number of the receiving application (e.g., 80=HTTP, 443=HTTPS, 22=SSH, 25=SMTP).
<u>Sequence Number</u>	32	Byte number of the first data byte in this segment. Used for ordering and reassembly. In SYN, this is the ISN (Initial Sequence Number).
<u>Acknowledgment Number</u>	32	Next byte the receiver expects. Confirms all bytes up to (ACK# - 1) received. Valid only when ACK flag is set.
<u>Data Offset (HLEN)</u>	4	TCP header length in 32-bit words. Minimum = 5 (20 bytes). Tells receiver where data begins.
Reserved	3	Must be zero. Reserved for future use.
<u>Control Flags</u>	9	URG, ACK, PSH, RST, SYN, FIN (+ ECE, CWR, NS). Critical for connection control! SYN=open, FIN=close, RST=abort, ACK=acknowledge.
<u>Window Size</u>	16	Number of bytes sender is willing to receive (flow control). Receiver advertises this; sender must not exceed it.
Checksum	16	Error detection over header + data + pseudo-header (src/dst IP, protocol, length).
Urgent Pointer	16	Points to urgent data when URG flag is set. Rarely used. Points to last byte of urgent data.
Options	0–320	Variable length. Common: MSS (Max Segment Size), Window Scaling, SACK (Selective ACK), Timestamps. Present only if Data Offset > 5.

Flag mnemonics — **SYN**=synchronize seq# **ACK**=acknowledge **FIN**=graceful close **RST**=abrupt reset **PSH**=push to app immediately

► UDP Header — User Datagram Protocol (Transport Layer)

💡 *UDP is connectionless, fast, and unreliable — no handshake, no ordering, no retransmission. Header is tiny: always exactly 8 bytes! Great for DNS, streaming, gaming, VoIP.*

Field	Size (bits)	Purpose / What it does
<u>Source Port</u>	16	Port number of the sending application. Optional in UDP — can be zero if no reply needed.
<u>Destination Port</u>	16	Port of the receiving application (e.g., 53=DNS, 67/68=DHCP, 123=NTP, 5353=mDNS).
<u>Length</u>	16	Total length of UDP header + data in bytes. Minimum value = 8 (header only).
Checksum	16	Optional in IPv4 (mandatory in IPv6). Covers header + data + pseudo-header. Zero means not used.

TCP vs UDP in one line: *TCP = reliable, ordered, heavy (20B min) | UDP = fast, unreliable, light (8B fixed) — pick based on whether you can tolerate loss.*

► Ethernet Frame — IEEE 802.3 (Data Link Layer)

💡 Ethernet is the most common LAN technology. It works at Layer 2 using MAC addresses. A frame wraps an IP packet. Standard Ethernet: 10/100/1000 Mbps (Gigabit).

Field	Size (bits)	Purpose / What it does
Preamble	56	7 bytes of alternating 1s and 0s (10101010...). Allows receiver's clock to synchronize with sender.
<u>SFD (Start Frame Delimiter)</u>	8	Always 10101011. Signals that the next byte is the start of the actual frame.
<u>Destination MAC</u>	48	6-byte MAC address of the intended recipient. Can be unicast, multicast (01:xx), or broadcast (FF:FF:FF:FF:FF:FF).
<u>Source MAC</u>	48	6-byte MAC address of the sender's NIC. Burned into hardware (though can be spoofed).
<u>EtherType / Length</u>	16	If ≥ 1536 (0x0600): identifies payload protocol (0x0800=IPv4, 0x0806=ARP, 0x86DD=IPv6, 0x8100=VLAN). If < 1536 : legacy 802.3 length field.
<u>Payload (Data)</u>	368–12000	The encapsulated packet (e.g., IP datagram). Minimum 46 bytes (padded if smaller to ensure collision detection). Max 1500 bytes normally.
<u>FCS (Frame Check Sequence)</u>	32	4-byte CRC-32 checksum over the entire frame. Receiver computes CRC; if mismatch → frame dropped silently.

► Token Ring Frame — IEEE 802.5 (Data Link Layer)

💡 *Token Ring uses a special 3-byte 'token' passed around the ring. A station can only transmit when it holds the token — eliminates collisions! Speeds: 4 or 16 Mbps. Largely replaced by Ethernet today.*

Field	Size (bits)	Purpose / What it does
<u>SD (Start Delimiter)</u>	8	Marks the beginning of the frame. Contains unique non-data signals (J and K symbols) to distinguish from data.
<u>AC (Access Control)</u>	8	Contains: Priority bits (3), Token bit (1 = free token), Monitor bit (1), Reservation bits (3). The Token bit is KEY — 0=token frame, 1=data frame.
<u>FC (Frame Control)</u>	8	Identifies frame type: LLC data frame (11xxxxxx) or MAC control frame (00xxxxxx). Controls how bridges process the frame.
<u>Destination Address</u>	48	6-byte MAC of the destination station on the ring.
<u>Source Address</u>	48	6-byte MAC of the transmitting station.
Data / Info	Variable	The payload — upper layer data (e.g., IP packet) or MAC management info.
FCS (Frame Check Sequence)	32	CRC-32 checksum for error detection, same concept as Ethernet.
<u>ED (End Delimiter)</u>	8	Marks end of frame. Contains E bit (error detected) and I bit (Intermediate frame — more frames follow).
<u>FS (Frame Status)</u>	8	Contains A bits (Address Recognized) and C bits (Frame Copied). Set by destination to confirm receipt. Unique to Token Ring!

Remember: *Token Ring = no collisions (deterministic), but token overhead adds latency. The FS field is unique — receiver modifies the frame in transit to confirm delivery!*

► SSL/TLS Record Header (Session/Transport Layer, OSI Layer 5/6)

💡 *SSL (deprecated) / TLS (current standard: TLS 1.3) encrypt traffic between client and server. Every TLS message is wrapped in a Record Layer header. HTTPS = HTTP + TLS.*

Field	Size (bits)	Purpose / What it does
<u>Content Type</u>	8	Identifies the type of data in the record. 20=ChangeCipherSpec, 21=Alert, 22=Handshake, 23=ApplicationData, 24=Heartbeat.
<u>Legacy Version</u>	16	Originally SSL/TLS version. In TLS 1.3 this is always 0x0303 (TLS 1.2) for compatibility. Actual version negotiated in handshake extension.
<u>Length</u>	16	Length of the record payload in bytes. Max = 16,384 bytes (2^{14}). Does NOT include the 5-byte record header itself.
Fragment / Payload	Variable	The actual content — handshake messages, encrypted application data, or alerts. Encrypted in all record types except initial handshake.

TLS Handshake Message Types (Content Type = 22):

Field	Size (bits)	Purpose / What it does
<u>ClientHello (1)</u>	—	Client sends: max TLS version, random nonce, supported cipher suites, extensions (SNI, supported_versions for TLS 1.3).
<u>ServerHello (2)</u>	—	Server responds: chosen version, random nonce, chosen cipher suite, session ID.
<u>Certificate (11)</u>	—	Server sends its X.509 certificate chain for authentication. Client verifies against trusted CAs.
ServerHelloDone (14)	—	TLS 1.2 only: server signals end of hello phase.
ClientKeyExchange (16)	—	TLS 1.2: client sends pre-master secret (RSA) or DH public key. In TLS 1.3, replaced by key_share extension.
<u>Finished (20)</u>	—	Both sides send a Finished message (HMAC of handshake transcript). Proves keys are correct. First encrypted message!

TLS 1.3 improvements: 0-RTT resumption | Removed weak ciphers (RC4, DES, 3DES) | Forward secrecy mandatory | Handshake reduced to 1-RTT | Encrypted certificates



Quick Comparison Table

Protocol	Layer	HDR Size	Addressing	Reliable?	Key Purpose
ICMP	Network(3)	8B min	None (uses IP)	No	Error reporting, ping, traceroute
IPv4	Network(3)	20–60 B	32-bit IP	No	Routing packets across networks; fragmentation
IPv6	Network(3)	40B fixed	128-bit IP	No	Next-gen routing; no fragmentation at routers
TCP	Transport(4)	20–60 B	Ports (16-bit)	Yes ✓	Reliable, ordered byte-stream delivery
UDP	Transport(4)	8B fixed	Ports (16-bit)	No	Fast, lightweight datagrams (DNS, streaming)
Ethernet	Data Link(2)	~26B	48-bit MAC	No (FCS)	LAN framing; CSMA/CD collision detection
Token Ring	Data Link(2)	~14B min	48-bit MAC	Token-based	Deterministic LAN; no collisions via token passing
SSL/TLS	Session(5/6)	5B record	Via TCP	Via TCP ✓	Encryption & authentication (HTTPS, etc.)

Good luck on your exam! 🚀